

Two-stage synthetic-to-real transfer learning for mammography report generation

MedGemma-4B + multimodal LoRA adapters, evaluated on DMID

STAGE 1 — SYNTHETIC PRETRAINING

DATA SOURCES



VinDr-Mammo

5,000 exams · 20k images
BI-RADS 0–5 · ACR A–D
no text reports



CBIS-DDSM

1,566 patients
mass · calcification
no text reports

REPORT SYNTHESIS



Label-to-report synthesis

Gemini 1.5 Flash · BI-RADS prompting + QA

100% QA pass

400 image+report

200+200 split



BI-RADS RAG

FAISS index
39 knowledge chunks
MiniLM-L6-v2 · top-1

PRETRAINING

synthetic pairs

LoRA transfer

MedGemma-4B backbone



SigLIP

vision encoder
frozen



Gemma-3

language model
frozen



LoRA

$r=16$, $a=32$
trainable

Stage 1 LoRA weights

pretrained on synthetic data

STAGE 2 — REAL DATA FINE-TUNING

TARGET DATASET



DMID Dataset

407 train

51 val

52 test

225 patient cases · 510 mammography images
Real radiologist reports · BI-RADS 1–5 · ACR A–D

FINE-TUNING

MedGemma-4B backbone



SigLIP

frozen



Gemma-3

frozen



LoRA

pretrained init
 $r=64$, $a=64$

Final two-stage model

INFERENCE PIPELINE

1

Mammogram
input

2

SigLIP
encoding

3

RAG
retrieval

4

Prompt
build

5

Generation

6

BI-RADS
report

LEGEND



frozen module



trainable (LoRA)

— data flow

--- weight transfer

--- retrieval context